Digits, place value and divisibility tests

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L1-NT2 Level 1 Foundations · Number theory

How to write a number in terms of its digits, and use this to explain the divisibility tests, find last digits of powers and solve digit puzzles.
By the end you can write a number in place value form and turn conditions on its digits into equations; use, and explain, the divisibility tests for 2, 3, 4, 5, 8, 9 and 11; find the last digit of a large power using cycles.
Prerequisites: Factors, multiples and primes (L1-NT1). **Contests:** JMC, Junior Kangaroo, JMO, AMC 8. **Time:** about 60 minutes.

1 Lesson

Take a two-digit number, reverse its digits and subtract the smaller number from the larger: $82 - 28 = 54$, $91 - 19 = 72$, $63 - 36 = 27$. Every answer is a multiple of 9. The reason is place value. A two-digit number with tens digit a and units digit b is $10a + b$, and its reverse is $10b + a$, so the difference is

$$(10a + b) - (10b + a) = 9a - 9b = 9(a - b),$$

which is 9 times a whole number.

1.1 Place value

We write $\overline{abc}$ for the three-digit number with digits a, b and c, so that

$$\overline{abc} = 100a + 10b + c.$$

The bar distinguishes it from the product $a \times b \times c$. Each digit is a whole number from 0 to 9, and the first is not 0.

Example 1.
A two-digit number is equal to 4 times the sum of its digits. Find all such numbers.

Solution.
Write the number as $10a + b$. The condition is $10a + b = 4(a + b) = 4a + 4b$. Taking $4a$ and b from both sides gives $6a = 3b$, so $b = 2a$.
Now use the limits on digits. The first digit a is at least 1, and $b = 2a$ is at most 9, so a is 1, 2, 3 or 4, giving 12, 24, 36 and 48. Check: the digits of 36 add to 9, and $4 \times 9 = 36$.

1.2 Tests that use the last digits

All the tests below rest on one fact: if two numbers are multiples of d, then so are their sum and their difference.

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Any number can be written as $10m + u$, where u is its units digit. Since $10 = 2 \times 5$, the part $10m$ is a multiple of 2 and of 5, so the number is a multiple of 2 exactly when u is, and of 5 exactly when u is 0 or 5. In the same way, $4728 = 4700 + 28$ and 4700 is a multiple of $100 = 4 \times 25$, so only the last two digits matter for 4; as $1000 = 8 \times 125$, only the last three matter for 8.

Tests for 2, 4, 5, 8 and 10.
A number is divisible by

- 2 if its last digit is even, 5 if its last digit is 0 or 5, and 10 if its last digit is 0;
- 4 if the number formed by its last two digits is divisible by 4;
- 8 if the number formed by its last three digits is divisible by 8.

Each test also works the other way round: if the last digits fail, the number fails.

1.3 Tests for 3 and 9

Each power of 10 is one more than a number made of 9s: $10 = 9 + 1$, $100 = 99 + 1$, $1000 = 999 + 1$, and so on. So for 4728,

$$4728 = 4 \times 1000 + 7 \times 100 + 2 \times 10 + 8 = (4 \times 999 + 7 \times 99 + 2 \times 9) + (4 + 7 + 2 + 8).$$

The first bracket is a multiple of 9 and the second is the digit sum, 21. So 4728 and 21 differ by a multiple of 9, and they leave the same remainder when divided by 9, namely 3. Since 9, 99, 999, ... are also multiples of 3, the same argument works for 3.

Theorem 1.
A number and the sum of its digits leave the same remainder when divided by 9, and the same remainder when divided by 3. In particular, a number is divisible by 9 (or by 3) if and only if its digit sum is.

Theorem 1 gives the remainder, not just a yes or no, and can be used twice: the digit sum of 98 765 432 is 44, whose digit sum is 8, so 98 765 432 leaves remainder 8 when divided by 9.

1.4 The test for 11

Now compare powers of 10 with multiples of 11: $10 = 11 - 1$, $100 = 99 + 1$ and $1000 = 1001 - 1$, where 11, $99 = 9 \times 11$ and $1001 = 7 \times 11 \times 13$ are all multiples of 11. So for a four-digit number,

$$\overline{abcd} = 1000a + 100b + 10c + d = (1001a + 99b + 11c) + (d - c + b - a).$$

The first bracket is a multiple of 11, and longer numbers work the same way.

Theorem 2.
Starting from the units digit, add and subtract the digits alternately. The number is divisible by 11 if and only if the result is divisible by 11. (The result may be 0 or negative; 0 counts as divisible by 11.)

For 2816 we get $6 - 1 + 8 - 2 = 11$, so 2816 is divisible by 11 (it is 11×256).

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1.5 Combining tests

A number is divisible by 6 if and only if it is divisible by 2 and by 3. The same holds for $12 = 3 \times 4$, $15 = 3 \times 5$, $18 = 2 \times 9$, $36 = 4 \times 9$, $45 = 5 \times 9$ and $72 = 8 \times 9$. A number divisible by both parts is divisible by their LCM (see L1-NT1), and in each case here the two parts have no common factor except 1, so their LCM is their product. Without that condition it fails: 12 is divisible by 4 and by 6, but not by 24.

Example 2.
The five-digit number $\overline{2A54B}$ is divisible by 72. Find it.

Solution.
Since $72 = 8 \times 9$, the number must be divisible by 8 and by 9. The test for 8 involves only B, so it comes first.
The last three digits $\overline{54B}$ must form a multiple of 8. The multiples of 8 near 540 are $8 \times 67 = 536$, $8 \times 68 = 544$ and $8 \times 69 = 552$, and only 544 lies between 540 and 549. So $B = 4$.
Now the digit sum $15 + A$ must be a multiple of 9. It lies between 15 and 24, so it equals 18 and $A = 3$. The number is 23544, and indeed $23544 = 72 \times 327$.

1.6 Last digits and cycles

The last digit of a sum or a product depends only on the last digits of the numbers. For example, 347×58 ends in 6, because $7 \times 8 = 56$ does. (Write $347 = 340 + 7$ and $58 = 50 + 8$ and multiply out: every part except 7×8 is a multiple of 10.) The powers of 2 are 2, 4, 8, 16, 32, 64, 128, 256, ..., with last digits 2, 4, 8, 6, 2, 4, 8, 6, ... Each last digit comes from doubling the previous one, so once 2 comes back, the same four digits repeat for ever.

last digit of a	remainder when k is divided by 4: 1	2	3	0
2	2	4	8	6
3	3	9	7	1
7	7	9	3	1
8	8	4	2	6

Cycles of last digits.
Powers of numbers ending in 2, 3, 7 or 8 repeat their last digits every 4 steps. The table gives the last digit of a^k. Powers of numbers ending in 4 end in 4, 6, 4, 6, ..., and those ending in 9 end in 9, 1, 9, 1, ... Powers of numbers ending in 0, 1, 5 or 6 always end in that digit.

For a cycle of length 4, divide the exponent by 4. A remainder of 1, 2 or 3 gives the first, second or third digit of the cycle, and a remainder of 0 gives the fourth.

Example 3.
What is the last digit of $8^{2027} + 9^{2027}$?

Solution.

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For 8^{2027}, the cycle is 8, 4, 2, 6. Since $2027 = 4 \times 506 + 3$, the remainder is 3, so 8^{2027} ends in the third digit of the cycle, which is 2. As a check, the exponent 3 leaves the same remainder, and $8^3 = 512$ ends in 2.
For 9^{2027}, the cycle is 9, 1, so odd powers such as 9^{2027} end in 9.
So the sum ends in the last digit of $2 + 9 = 11$, which is 1.

1.7 Digit puzzles

In a digit puzzle, the place value form turns each condition into an equation, the divisibility tests handle conditions about multiples, and the limits on digits (each from 0 to 9, the first never 0) leave a short list of cases. Check each case against the whole statement: simplifying can lose a condition, such as a reversed number also having three digits.

Example 4.
A three-digit number is divisible by 11. Reversing its digits gives another three-digit number, which is 594 less than the original. Find all possible original numbers.

Solution.
Write the number as $\overline{abc} = 100a + 10b + c$. Its reverse $100c + 10b + a$ has three digits, so c is at least 1. The difference is

$$(100a + 10b + c) - (100c + 10b + a) = 99a - 99c = 99(a - c),$$

so $99(a - c) = 594$ and $a - c = 6$. The middle digit has cancelled, so we find it with the test for 11.
By Theorem 2, $c - b + a$ is divisible by 11. It lies between -9 and 18, so it is 0 or 11, that is, $b = a + c$ or $b = a + c - 11$.
Now $a = c + 6$ with $c \geq 1$ and $a \leq 9$, so c is 1, 2 or 3. If $c = 1$ then $a = 7$ and b is 8 or -3, so $b = 8$, giving 781. If $c = 2$ then $a = 8$ and b is 10 or -1, neither a digit. If $c = 3$ then $a = 9$ and b is 12 or 1, so $b = 1$, giving 913.
So the original number is 781 or 913. Checking: $781 = 11 \times 71$ and $781 - 187 = 594$; $913 = 11 \times 83$ and $913 - 319 = 594$.

Common traps.

- The digit-sum test works for 3 and 9 only. The digit sum of 2799 is 27, but 2799 is not divisible by 27.
- The test for 4 uses the last two digits as a number. The number 34 ends in 4 but is not a multiple of 4.
- An exponent that is a multiple of 4 gives the *fourth* digit of a cycle of length 4: 3^{40} ends in 1, not 3.
- A first digit is never 0. The reverse of 120 is not a three-digit number.

1.8 How to spot this in a contest

1. Missing digits and a divisor: split the divisor into parts with no common factor, use the last-digit tests first, then the digit sum or the test for 11.

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2. Reversed digits, digit sums, or "a number equals so many times the sum of its digits": write the number in place value form and simplify.
3. The last digit of a large power: find the cycle, and divide the exponent by the length of the cycle.
4. Letters standing for digits: look at units digits, then sizes (a first digit is never 0, and a product cannot have too many digits), then divisibility tests.
5. Repeated blocks of digits come from products with 11, 101, 111 or 1001. For example $\overline{abab} = \overline{ab} \times 101$ and $\overline{aaa} = a \times 111 = a \times 3 \times 37$.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ What is the remainder when the ten-digit number 2026202620 is divided by 9?

Problem 2 ⋆ What is the units digit of $2^{30} + 3^{30}$?

Problem 3 ⋆⋆ How many four-digit numbers of the form $\overline{7A2B}$ are divisible by 12?

Problem 4 ⋆⋆ How many five-digit numbers that use each of the digits 1, 2, 3, 4 and 5 exactly once are divisible by 24?

Problem 5 ⋆⋆⋆ Explain why no perfect square has digit sum 2027.

Problem 6 ⋆⋆⋆ In the sum $\overline{ABC} + \overline{ABC} + \overline{ABC} = \overline{BBB}$, the letters A, B and C stand for three different digits, and A is not 0. Find the three-digit number $\overline{ABC}$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT JMC 2018 Question 11** ⋆
   Write the big number in blocks of three digits using place value.
2. **AMC 8 2024 Problem 1** ⋆
   Only the units digits matter.
3. **UKMT JMC 2017 Question 10** ⋆
   Split 45 into 5 and 9 and use two tests.
4. **AMC 8 2012 Problem 12** ⋆⋆
   Powers of 13 end in the same digits as powers of 3.
5. **AMC 8 2000 Problem 14** ⋆⋆
   Powers of numbers ending in 9 have a cycle of length 2.

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6. **UKMT JMC 2019 Question 10** ⋆⋆
   Fewer digits means a smaller number; then decide which digit should go first.

7. **AMC 8 2017 Problem 7** ⋆⋆
   Write Z in place value form as two blocks of three digits.

8. **AMC 8 2014 Problem 22** ⋆⋆
   Write the number as $10a + b$ and simplify.

9. **UKMT JMC 2019 Question 22** ⋆⋆⋆
   Nine times TRAP still has four digits, which limits T; then look at units digits.

10. **AMC 8 2020 Problem 19** ⋆⋆⋆
    Split 15 into 3 and 5, and remember that the last digit of a five-digit flippy number equals its first.

11. **UKMT JMC 2015 Question 24** ⋆⋆⋆
    Split 45 into 5 and 9; the last digit of a palindrome is also its first digit.

12. **UKMT JMO 2018 Question B3** ⋆⋆⋆
    Use the test for 5 first, then the test for 8, then the test for 3.

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4 Hints

1. A number and its digit sum leave the same remainder when divided by 9.
2. Both cycles have length 4. What is the remainder when 30 is divided by 4?
3. Use $12 = 3 \times 4$. The test for 4 tells you which values B can take; then handle each one with the test for 3.
4. Split 24 into 3 and 8. One of the two tests is passed by every such number, so only the other one matters.
5. Use Theorem 1 with the number 3. What remainders can a square leave when divided by 3?
6. Write $\overline{BBB}$ as $B \times 111$ and factorise 111.

5 Solutions

Problem 1

What is the remainder when the ten-digit number 2026202620 is divided by 9?

By Theorem 1, the number leaves the same remainder as its digit sum. The digits are 2, 0, 2, 6, 2, 0, 2, 6, 2, 0, with sum $10 + 10 + 2 = 22$. Since $22 = 2 \times 9 + 4$, the remainder is 4.

Answer: 4

Problem 2

What is the units digit of $2^{30} + 3^{30}$*?*

The last digits of the powers of 2 run 2, 4, 8, 6 and repeat, and those of the powers of 3 run 3, 9, 7, 1 and repeat. Since $30 = 4 \times 7 + 2$, both 2^{30} and 3^{30} are at the second place in their cycles, so 2^{30} ends in 4 and 3^{30} ends in 9. The sum ends in the last digit of $4 + 9 = 13$, so the units digit is 3.

Answer: 3

Problem 3

How many four-digit numbers of the form $\overline{7A2B}$ *are divisible by 12?*

Since $12 = 3 \times 4$ and 3 and 4 have no common factor except 1, the number must be divisible by 4 and by 3.

For 4, the last two digits $\overline{2B}$ must form a multiple of 4. The multiples of 4 from 20 to 29 are 20, 24 and 28, so B is 0, 4 or 8.

For 3, the digit sum $7 + A + 2 + B = 9 + A + B$ must be a multiple of 3, which means $A + B$ must be a multiple of 3. If $B = 0$, then A is 0, 3, 6 or 9, which is 4 numbers. If $B = 4$, then $A + 4$ is a multiple of 3, so A is 2, 5 or 8, which is 3 numbers. If $B = 8$, then $A + 8$ is a multiple of 3, so A is 1, 4 or 7, which is 3 numbers. In total there are $4 + 3 + 3 = 10$ such numbers.

Answer: 10

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Problem 4

How many five-digit numbers that use each of the digits 1, 2, 3, 4 and 5 exactly once are divisible by 24?

Since $24 = 3 \times 8$ and 3 and 8 have no common factor except 1, the number must be divisible by 3 and by 8.

The digit sum is always $1 + 2 + 3 + 4 + 5 = 15$, which is a multiple of 3, so every such number is divisible by 3. Hence we only need the test for 8: the last three digits must form a multiple of 8.

A multiple of 8 is also a multiple of 4, so the last two digits must form a multiple of 4. Using two different digits from 1 to 5, the only possibilities are 12, 24, 32 and 52. For each, the hundreds digit is one of the three digits not yet used, and we test the last three digits for 8:

with 12 we get 312, 412 and 512, and $312 = 8 \times 39$ and $512 = 8 \times 64$ work, while 412 does not; with 24 we get 124, 324 and 524, none of which works; with 32 we get 132, 432 and 532, and only $432 = 8 \times 54$ works; with 52 we get 152, 352 and 452, and $152 = 8 \times 19$ and $352 = 8 \times 44$ work. So there are 5 possible endings: 152, 312, 352, 432 and 512.

For each ending, the two digits not used in it fill the first two places in $2 \times 1 = 2$ ways. So there are $5 \times 2 = 10$ such numbers, for example 43152 and 34152.

Answer: 10

Problem 5

Explain why no perfect square has digit sum 2027.

By Theorem 1, a number with digit sum 2027 leaves the same remainder on division by 3 as 2027 does. Since $2027 = 3 \times 675 + 2$, that remainder is 2. So it is enough to show that a square never leaves remainder 2 when divided by 3.

Every whole number n is of the form $3k$, $3k + 1$ or $3k + 2$ for some whole number k. In the first case $n^2 = 9k^2 = 3 \times 3k^2$, which leaves remainder 0. In the second case $n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$, which leaves remainder 1. In the third case $n^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$, which also leaves remainder 1.

So a square leaves remainder 0 or 1 when divided by 3, never 2, and hence no square has digit sum 2027.

Problem 6

In the sum $\overline{ABC} + \overline{ABC} + \overline{ABC} = \overline{BBB}$, the letters A, B and C stand for three different digits, and A is not 0. Find the three-digit number $\overline{ABC}$.

The left side is $3 \times \overline{ABC}$, and $\overline{BBB} = 100B + 10B + B = 111B$. Since $111 = 3 \times 37$, the equation becomes $3 \times \overline{ABC} = 3 \times 37 \times B$, so $\overline{ABC} = 37B$.

Now B is a digit from 1 to 9 (if $B = 0$ then $\overline{ABC} = 0$), and the middle digit of $37B$ must be B itself. The multiples of 37 are 37, 74, 111, 148, 185, 222, 259, 296 and 333 for $B = 1, 2, \ldots, 9$. The first two have only two digits, and 111, 222 and 333 do not have three different digits. Of the rest, only 148 has middle digit equal to $B = 4$: for 185 the middle digit is 8, not 5; for 259 it is 5, not 7; and for 296 it is 9, not 8.

So $\overline{ABC} = 148$, and indeed $148 + 148 + 148 = 444$.

Answer: 148

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The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.